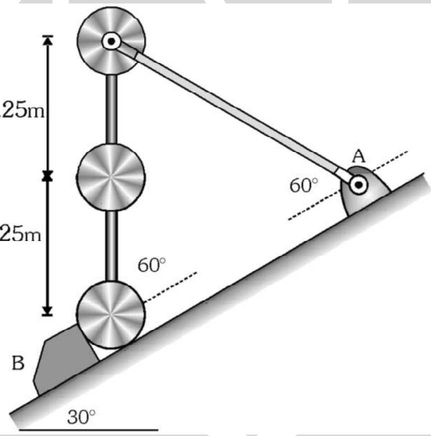
TOPIC : Work, Power, Energy & Circular Motion

SUBJECT: PHYSICS **DATE: 19-03-2024**

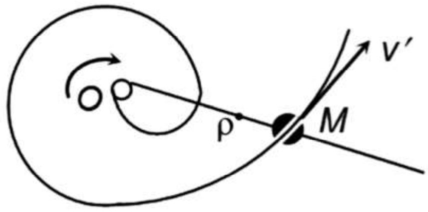
SINGLE CORRECT TYPE (+3, –1)

1. The three identical spheres, each of mass m, are supported in the vertical plane on the 30º incline. The spheres are welded to the two connecting rods of negligible mass. The upper rod, also of negligible mass, is pivoted freely to the upper sphere and to the bracket at A. The upper sphere can rotate freely about pivote. If the stop at B is suddenly removed, determine the velocity v with which the upper sphere hits the incline. Neglet friction at all the places.



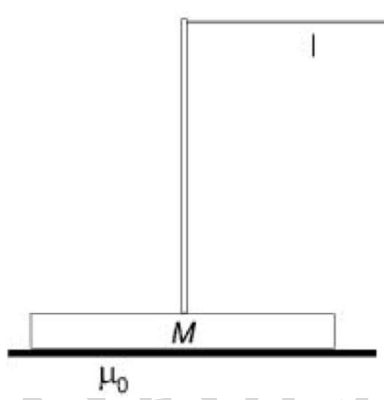
(A) 5.5 m/s (B) 4.20 m/s (C) 8.0 m/s (D) 2.0 m/s

2. A plane spiral made of stiff smooth wire is rotated with a constant angular velocity ω in a horizontal plane about the fixed vertical axis O. A small sleeve M slides along that spiral without friction. Find its velocity v' relative to the spiral as a function of the distance ρ from the rotation axis O (The initial velocity of the sleeve is equal to v_0)



(A) $\sqrt{v_o^{\,2}+\rho^2\omega^2}$ (B) $\sqrt{v_0^2+\frac{\left(\rho^2\omega^2\right)}{2}}$ (C) $v_o+\rho\omega$ (D) $\frac{v_o^{\,2}+\rho^2\omega^2}{2}$

3. A pointlike ball of mass m is tied to the end of a string, which is attached to the top of a thin vertical rod. The rod is fixed to the middle of a block of mass M lying at rest on a horizontal plane. The pendulum is displaced to a horizontal position and released from rest.

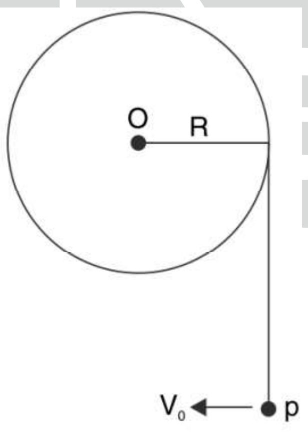


If the coefficient of static friction between the block and the ground is μ_s, what angle will the string create with the vertical rod at the time instant when the block starts to slide? ($M = 2kg$, $m = 1\ kg$, μ_s=0.2.) [Neglect any toppling)

(A) $\sin^{-1}\sqrt{0.98}$ (B) $\sin^{-1}(0.6)$ (C) $\sin^{-1}\sqrt{0.28}$ (D) $\sin^{-1}(0.78)$

ONE OR MORE THAN ONE OPTION(S) CORRECT: (+4, –1)

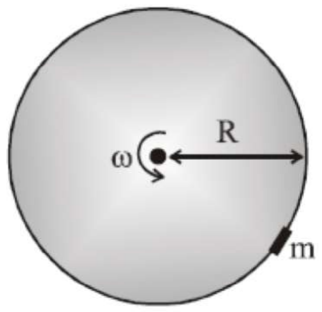
4. A light thread is tightly wrapped around a fixed disc of radius R. A particle of mass m is tied to the end P of the thread and the vertically hanging part of the string has length πR. The particle is imparted a horizontal velocity $V = \sqrt{\frac{4\pi gR}{3}}$. The string wraps around the disc as the particle moves up. At the instant the velocity of the particle makes an angle of $\theta = 60^o$ with horizontal.



(A) speed of particle is $\sqrt{3}gR$ (B) speed of particle is $\sqrt{3gR}$

(C) tension in string is $mg\sqrt{3}$ (D) tension in string is $mg\left(\sqrt{3}+\frac{1}{2}\right)$

5. A rough vertical circular ring of radius R, carrying a bead of mass m, rotates in its own plane about its centre with uniform angular velocity ω. The coefficient of friction between ring and bead is μ. Choose the correct statement(s).



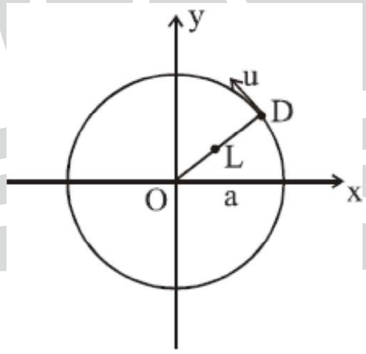
(A) Minimum angular speed of ring such that bead does not slip on ring is $\sqrt{\frac{g}{\mu R}\sqrt{1+\mu^2}}$

(B) Minimum angular speed of ring such that bead does not slip on ring is $\sqrt{\frac{g}{\mu R}}$

(C) If bead does not slip on ring, maximum frictional force experienced by it is equal to mg

(D) If bead does not slip on ring, minimum normal reaction force experienced by it is equal to $\left(\left|m\omega^2 R - mg\right|\right)$

6. Poor Dinanath (D) is thrown into a circular arena of radius a containing a lion (L). Initially the lion is at the centre O of the arena while Dinanath is at $(a, 0)$. Dinanath decides to run at his maximum speed u around the perimeter. The lion responds by running at its maximum speed v in such a way that it remains on the (moving) radius OD. Choose the correct statement(s).



(A) During the chase, distance between lion and Dinanath at time t is $\left[a-\frac{va}{u}\sin\left(\frac{ut}{a}\right)\right]$

(B) If a =10 m, u = 5 m/s and v =10 m/s then lion catches Dinanath after $\frac{\pi}{3}s$

(C) Path followed by lion is circle

(D) Path followed by lion is hyperbola

7. A bead of mass m can slide on a smooth circular wire of radius a, which is fixed in a vertical plane. The bead is connected to the highest point of the wire by a light spring of natural length $3a/2$ and spring constant k. P is lowest point on ring. Choose the CORRECT statement(s).

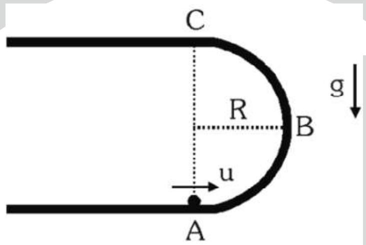
(A) For bead, P is stable equilibrium position if $k = \frac{2mg}{a}$

(B) For bead, P is unstable equilibrium position if $k = \frac{2mg}{a}$

(C) For bead, P is stable equilibrium position if $k = \frac{5mg}{a}$

(D) For bead, P is unstable equilibrium position if $k = \frac{5mg}{a}$

8. Figure shows a smooth fixed track with two straight parts connected by a semicircular arc ABC. The shown track is placed in a vertical plane. A small ball is given an initial speed 'u' along the track as shown. The ball loses contact with the track somewhere in the part BC and collides with the track at A during subsequent motion. Now answer the following questions.



(A) the value of 'u' is $\sqrt{\frac{7}{2}Rg}$

(B) time after which it collides at 'A' after losing the contact with the shown track, is $\sqrt{\frac{6R}{g}}$

(C) the value of 'u' is $\sqrt{\frac{9}{2}Rg}$

(D) The time after which it collides at 'A' after losing the contact with the shown track, is $\sqrt{\frac{3R}{g}}$

9. A worker wishes to pile up sand on to a circular area of radius R. No sand is to spill on to the surrounding area. If μ_s is the coefficient of static friction between sand particles, then choose the correct statement(s).

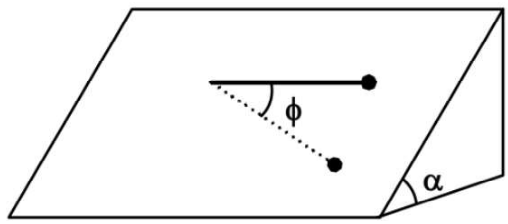
(A) The greatest height of sand pile that can be created is $\mu_s R$.

(B) The greatest height of sand pile that can be created is R / μ_s.

(C) Minimum work required to create greatest height sand pile is $\frac{\pi}{12}\mu_s^2 R^4 \rho g$. Where ρ is the density of sand volume.

(D) Sand pile will be in stable equilibrium.

10. One end of a thread of length I = 1m is attached to an inclined plane of angle of elevation of $\alpha = 30^o$ .A point-like body of mass $m = 1kg$ is attached to the other end of the thread as shown in the figure. The body is released without initial speed, such that the thread is tight and horizontal. The coefficient of friction between the slope and the body is $\mu = 0.2$.



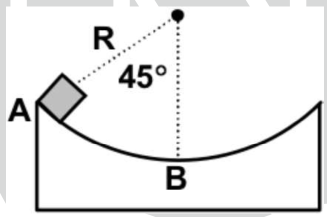
(A) The tension in the thread when the angle between the thread and the horizontal is ϕ is $T = (3\sin\phi\sin\alpha - 2\mu\phi\cos\alpha)mg$

(B) The tension in the string will be maximum when $\phi = 90^o$.

(C) The tension in the string will be maximum when $\phi = \cos^{-1}\left(\dfrac{2\mu}{3\tan\alpha}\right)$

(D) The maximum speed of the particle will be when $\phi = \cos^{-1}(\mu\cot\alpha)$

11. A small rubber eraser is placed at one edge of a quarter-circle-shaped track of radius R that lies in a vertical plane and has its axis of symmetry vertical (see figure); it is then released. The coefficient of friction between the eraser and the surface of the track is $\mu = 0.6$.



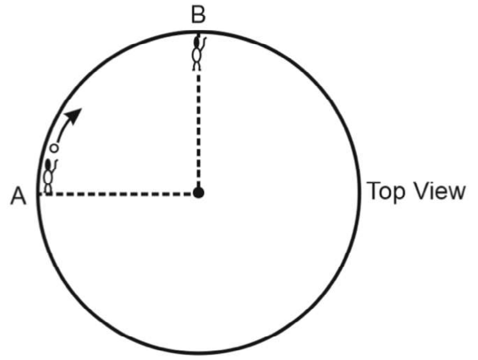
(A) If the particle slides from A to B work done by frictional force will be $\dfrac{0.6mgR}{\sqrt{2}}$.

(B) If the particle slides from A to B work done by frictional force will be greater than $\dfrac{0.6mg\pi R}{4\sqrt{2}}$.

(C) The particle will never be able to go from A to B.

(D) If μ is $\sqrt{2}$ the particle will not begin to slide.

12. Two friends A and B (each weighing 40 kg) are sitting in a frictionless well having vertical circular smooth wall. They subtend angle 90° at the centre of well. 'A' rolls a ball of mass 40 kg in clockwise direction on the platform towards B along the wall of the well which B catches. Then similarly B rolls the ball towards 'A' anticlockwise and A catches it. The ball has a fixed speed of 5 m/s. (Neglect catching and throwing time.)



(A) The total impulse by the wall of well on whole system by the time ball travels from A to B is 200 *kg-m/s*
(B) The total impulse by the wall of well on whole system by the time ball travels from A to B is 400 *kg-m/s*
(C) The angular displacement of A till he meets B is 120º
(D) The angular displacement of A till he meets B is 150º

13. A particle of mass m moves in a certain plane P due to a force F whose magnitude is constant and whose vector rotates in that plane with a constant angular velocity 'w'. Assuming the particle to be stationary at $t = 0$,

(A) The speed of the particle at any time 't' is $v(t) = \frac{2F}{m\omega}\left|\sin\left(\frac{\omega t}{2}\right)\right|$

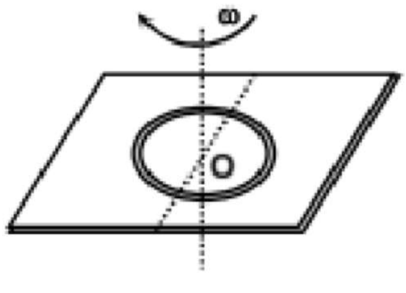
(B) The speed of the particle at any time 't' is $v(t) = \frac{F}{m\omega}\left|\sin\left(\frac{\omega t}{2}\right)\right|$

(C) The distance covered by the particle between two successive stops is $\Delta s = \frac{8F}{mw^2}$

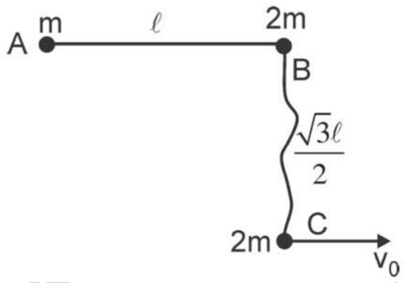
(D) Mean speed between two successive stops is $< v >= \frac{4F}{\pi m w}$

NUMERICALS TYPE: (+4, –1)

14. A uniform ring of mass m is placed on a rough horizontal fixed surface as shown in the figure. The coefficient of friction between the left part of the ring and left part of the horizontal surface is $\mu_1 = 0.6\pi$ and between right half and the surface is $\mu_2 = 0.2\pi$. At the instant shown, now the ring has been imparted an angular velocity in clockwise sense in the figure shown. At this moment magnitude of acceleration of centre O of the ring (in m/s^2) is (take $g = 10\,m/s^2$)

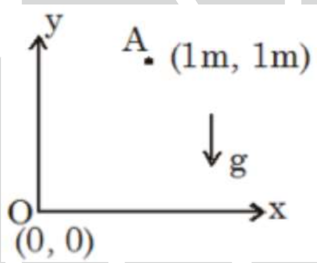


15. Three small balls A, B and C of masses m, $2m$ and $2m$ respectively are placed on a smooth horizontal surface. Balls A and B as well as B and C are connected by light inextensible strings each of equal length ℓ. The string connecting balls A and B is taut. The initial distance between the balls B and C is $\frac{\ell\sqrt{3}}{2}$ and BC is perpendicular to AB. The ball 'C' is given a horizontal velocity $v_0 = 23\,m/s$ parallel to AB as shown in the figure. Find the initial velocity of ball A (in m/s) just after string BC becomes taut.



Passage for Nos. (16 to 17)

Answer the following by appropriately matching the lists based on the information given in the paragraph A rough track connects point O ($0m$, $0m$) to point A($1m$, $1m$). External force F always tangential to path pulls object of mass 2 kg from point $O(0m, 0m)$ to point $A(1m, 1m)$ very slowly. Coefficient of friction between block and track is 0.5



In List-I some physical quantity regarding motion of block is mentioned while List-II has its corresponding numerical value.

List - I	**List - II**
(I) Work done by frictional force on block magnitude wise in Joules	(P) $10\sqrt{2}$
(II) Normal reaction force between block and track at $x = \frac{1}{2}m$ in Newton	(Q) $20\sqrt{\frac{2}{3}}$
(III) Work done by external force F on block magnitude wise in Joules	(R) 10
(IV) Net reaction force between block and track at $x = \frac{3}{4}m$ Newton	(S) $20\sqrt{\frac{5}{13}}$
	(T) $5\sqrt{15}$
	(U) 30

16. If equation of track is $y = x^2$ then choose the correct combination from List I & List II.

(A) I $\rightarrow$ S; II $\rightarrow$ P; III $\rightarrow$ Q; IV $\rightarrow$ T
(B) I $\rightarrow$ U; II $\rightarrow$ P; III $\rightarrow$ T; IV $\rightarrow$ S
(C) I $\rightarrow$ Q; II $\rightarrow$ U; III $\rightarrow$ S; IV $\rightarrow$ T
(D) I $\rightarrow$ R; II $\rightarrow$ P; III $\rightarrow$ U; IV $\rightarrow$ S

17. If equation of track is $y^2 = x$ then choose the correct combination from List I & List II.

(A) I $\to$ R; II $\to$ Q; III $\to$ U; IV $\to$ T

(B) I $\to$ S; II $\to$ Q; III $\to$ T; IV $\to$ U

(C) I $\to$ P; II $\to$ Q; III $\to$ R; IV $\to$ S

(D) I $\to$ U; II $\to$ Q; III $\to$ T; IV $\to$ R

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